

# ABHISHEK KUMAR SINGH

Site Reliability Engineer | Cloud Architect

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Site Reliability Engineer specialized in building high-availability, self-healing infrastructure with a focus on SLO-driven operations and Toil reduction. Architected event-driven AIOps frameworks achieving 97% MTTR reduction and engineered Zero-Trust Kubernetes environments on AWS.

Available for Remote Roles (Singapore / UK / Global). Strategic IST overlap with SGT (+2.5 hrs) and GMT (+5.5 hrs) for seamless collaboration.

## TECHNICAL SKILLS

- **Infrastructure:** AWS (EKS, EC2, VPC, IAM, S3, EventBridge), Terraform (IaC), Kubernetes, Docker, Linux Admin, Disaster Recovery and Chaos Engineering.
- **Security & DevSecOps Engineering:** Snyk, Trivy, SonarQube, OIDC Federation, IRSA, Zero-Trust Networking, Static & Dynamic Analysis (Slither, Certora).
- **AIOps & Observability:** Monitoring Four Golden Signals (Latency, Traffic, Errors, Saturation) via Prometheus, Grafana and CloudWatch Anomaly Detection; Incident Management with AlertManager.
- **Automation & CI/CD:** GitHub Actions, Jenkins, Git, Helm, Python, Shell Scripting.
- **Data Science & Analytics:** Statistical Analysis, Predictive Modeling, Pandas, Scikit-Learn (IIIT Bangalore Certified).

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## ENGINEERING PROJECTS

### Self-Healing EKS Remediation Engine (AIOps/AWS)

AIOps · AWS Lambda · EventBridge · Python · CloudWatch | 2025 | [GitHub Link](#)

- Reduced MTTR by 97% (from 20m to <60s) by architecting an event-driven AIOps framework that automates Kubernetes incident response and node recovery workflows.
- Optimized observability via CloudWatch Anomaly Detection and Prometheus AlertManager, cutting false-positives by 40% while maintaining 99.9% availability during chaos testing.

### Infrastructure EKS Provisioner

Terraform · AWS EKS · VPC · IAM · IRSA | 2025 | [GitHub Link](#)

- Architected modular Terraform to provision a production-grade, multi-AZ EKS cluster with private subnets and NAT gateways in under 12 minutes.
- Enforced Zero-Trust security via IRSA, successfully eliminating static credentials and passing simulated penetration checks with zero exposed attack surface.

### DevSecOps CI/CD Pipeline

GitHub Actions · Trivy · Docker · Snyk · ECR · OIDC | 2025

- Engineered a "Shift-Left" pipeline with integrated Trivy/Snyk gating, achieving a 100% block rate on critical CVEs during production-ready testing.
- Automated secure deployments via OIDC federation and multi-stage Docker builds, reducing container image sizes by 60% and pipeline runtime by 35%.

### Scalable Data Management API

Python · FastAPI · MongoDB · Docker | 2024 | [GitHub Link](#)

- Developed an asynchronous REST API sustaining 500+ concurrent requests with zero blocking I/O, utilizing Pydantic to ensure 0% malformed-record persistence.
- Optimized container footprint to under 80MB via multi-stage builds, significantly lowering ECR storage costs and cutting deployment cold-start times.

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## EMPLOYMENT HISTORY

### Site Reliability Engineer (Independent Projects)

Remote | Jul 2025 – Present

- Mitigated operational risk by architecting multi-AZ AWS environments with automated failover and Zero-Trust identity management (IRSA).
- Reduced operational toil by 40% through the development of Boto3 automation scripts that replaced manual infrastructure auditing and node maintenance.
- Delivered automated AIOps remediation solutions using AWS Lambda and EventBridge, reducing Mean Time to Recovery (MTTR) by 97% through self-healing workflows.
- Standardized secure-by-design CI/CD pipelines with integrated vulnerability gating (Trivy, Snyk), achieving 100% security compliance across all containerized deployments.

### Graduate Administrative Assistant

Troy University, Alabama, USA | May 2024 – Jun 2025

- Optimized and administered the WC Online Portal database to streamline scheduling and payroll workflows, ensuring 100% data integrity for university-wide tutor operations.
- Spearheaded technical workshops on generative AI and emerging tech trends, translating complex software concepts into actionable training for a diverse community of students and faculty.
- Provided advanced technical mentorship to Computer Science undergraduates, performing code reviews and debugging sessions to reinforce secure programming and algorithmic efficiency.

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## EDUCATION

### Master of Science in Computer Science (Specialization in Cybersecurity)

Jan 2024 – Jun 2025

Troy University, Alabama, USA

Research Thesis: Smart Contract Measurement Methodology - Extracting & Analysing KPIs. Developed a taxonomy for evaluating smart contract robustness using Static Analysis (Slither), Dynamic Profiling (Tenderly), and Formal Verification (Certora).

### Advanced Certification in Data Science

Mar 2023 – Nov 2023

IIIT Bangalore, India

GPA: 3.7/4.0 • Machine Learning, Predictive Analysis, Python for Data Analysis

### Bachelor of Science (Electronics)

Jun 2018 – Sep 2021

Guru Ghasidas Central University, India

Capstone: Greenhouse Control & Management System using PIC Microcontrollers and sensor network.

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## CERTIFICATIONS & ACHIEVEMENTS

- **HashiCorp Certified: Terraform Associate (004)** – In Progress
- **Winner (1<sup>st</sup> Position), TroyHack 2024** – Secured 1<sup>st</sup> position in a 24-hour hackathon by rapid-prototyping a 3D simulation game on an education theme using Unity & C#
- **International Olympiad of Mathematics** – 1<sup>st</sup> Position in School (2016), 5<sup>th</sup> Rank in State (2017)
- **IIIT Bangalore Data Science Certification** – GPA 3.7/4.0